



"Your Partner in Innovative Refrigeration and Advanced Testing Solutions."

SRI EASWARI SCIENTIFIC SOLUTION PVT LTD
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INTRODUCTION

Welcome to Sri Easwari Scientific Solution Pvt Ltd (SESS)

Established in 2010, Sri Easwari Scientific Solution Pvt Ltd (SESS) has become a trusted name in the field of environmental testing solutions, offering cutting-edge climatic test chambers, refrigeration systems, and customized control solutions. Headquartered in Chennai, SESS is driven by a commitment to innovation, quality, and customer satisfaction.

Our Expertise

Environmental Test Chambers

- Thermal Cycling Chambers
- Humidity and Temperature Chambers
- Walk-in Stability Chambers
- Custom Test Chambers

Customized Solutions

Specialized in developing bespoke environmental and stability testing solutions tailored to unique industry requirements.

MEP Consulting & Design

Comprehensive services in Mechanical, Electrical, and Plumbing (MEP) projects for industrial and commercial applications.

Industries We Serve

- Aerospace and Defense
- Automotive
- Electronics Manufacturing
- Pharmaceutical Stability Testing
- Research and Development

Why Choose SESS?

Innovation-Driven:

We use state-of-the-art technologies to meet evolving testing standards.

Customer-Centric Approach:

Solutions designed to deliver exceptional value and long-term support.

Expert Team:

Skilled professionals offering reliable technical support, installation, and maintenance.

Organizational Excellence

Production Teams:

Mechanical, Electrical, Instrumentation, and Refrigeration experts.

Service Teams:

Dedicated professionals ensuring seamless post-sale support.

Sales and Coordination Teams:

Focused on delivering outstanding client service.

SESS GROUP OF COMPANIES

Sri Easwari Scientific Solution

Focus: Technical support and service solutions.

Expertise:

Comprehensive support for all our group companies, specializing in the maintenance and servicing of environmental test chambers and related equipment.

Sri Easwari Scientific Solution Pvt Ltd

Focus: Industrial Climatic Testing Chambers

Manufacturing:

Advanced thermal and environmental test chambers, including Thermal Shock Test Chambers, Humidity Chambers, and Temperature Cycling Chambers.

Industries Served: Aerospace, Defense, Automotive, Electronics, and R&D Labs.



GEOCLIMATIC TECHNOLOGIES PVT LTD
Est. 1999

Focus: Custom-Engineered Special Projects

Pharmaceutical Testing Chambers:

Tailored solutions for stability and climatic testing in the pharma sector.

Customized Projects:

Developing bespoke environmental testing solutions to meet unique customer requirements.

Srisa Design Contracting Engineers



Focus: MEP Consulting & Design Services

MEP Projects: Comprehensive consulting and contracting services for Mechanical, Electrical, and Plumbing projects.

Design Support: Providing end-to-end design solutions and project management for industrial and commercial sectors.

Our Commitment

Innovation & Excellence:

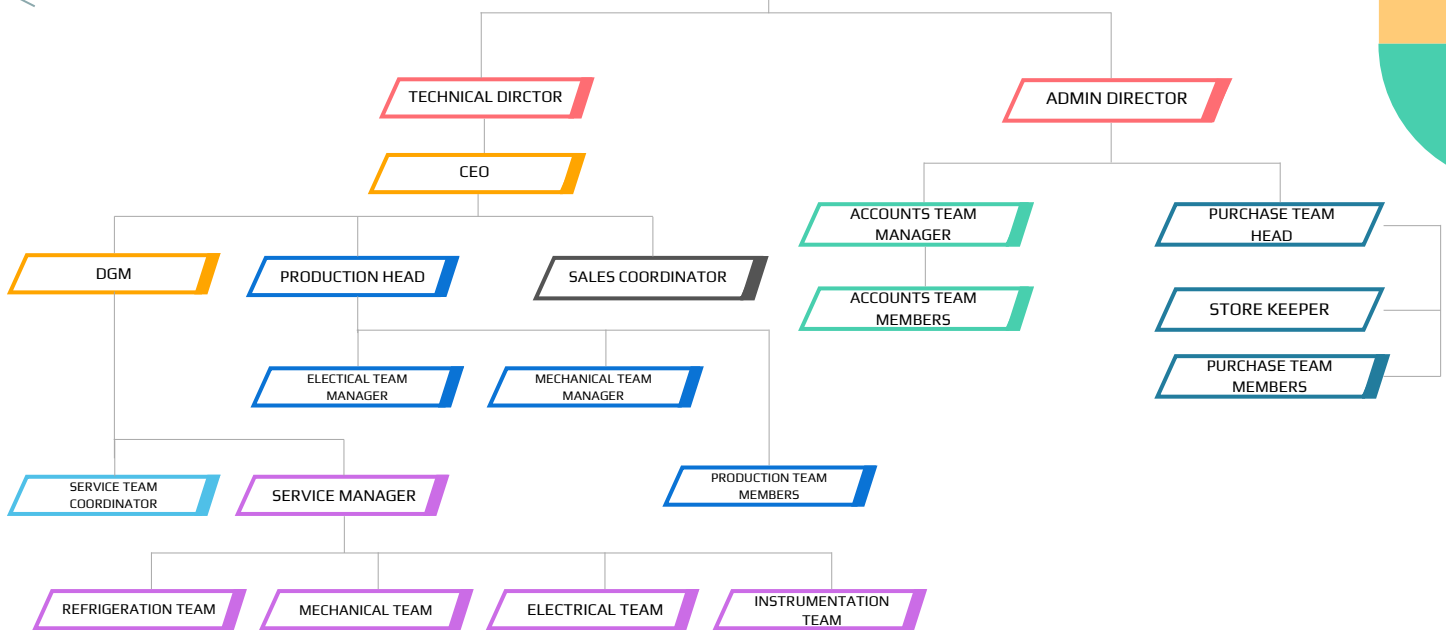
Driving industry standards with cutting-edge technology and tailored solutions.

Customer-Centric Approach:

Focused on delivering exceptional value and support to our clients.

SESS ORGANIZATION STRUCTURE

Sri Easwari Scientific Solution Pvt Ltd



OUR ESTEEMED CLIENTS



More than 100+ Clients..

OUR PRODUCTS



Climatic Test Chambers

- For environmental testing across temperature, humidity, and other conditions.

● **Applications:** Automotive, Electronics, Aerospace.

Thermal Shock Test Chambers

For rapid temperature cycling to test material durability.

● **Applications:** Aerospace, Defense, Manufacturing.



Salt Spray Test Chambers

- For corrosion resistance testing

● **Applications:** Automotive, Electronics, Military.

Rain Test Chambers

To simulate rain and moisture exposure.

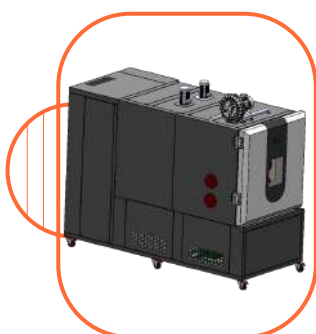
● **Applications:** Automotive, Consumer Electronics.



Walk-In Stability Chambers

- For large-scale environmental testing.

● **Applications:** Pharmaceutical, Electronics.



🔋 *1000-Liter Capacity*

Battery Test Chamber 🔋

Vibration Combined Climatic Test Chamber



NEW PROJECTS ACHIEVEMENTS

Advanced Cryogenic High-Speed Thermal Cyclic Test Chamber for Aerospace Applications

We have supplied a Liquid Nitrogen-Based Thermal Cyclic Test Chamber to **LPSC ISRO**, designed for advanced aerospace testing.

Key Features:

Temperature Range: +180°C to -180°C

Ramp Rate: 60°C per minute

This state-of-the-art test chamber ensures precision and reliability for critical aerospace applications.



Thermal Cyclic Test Chamber Supplied to BrahMos Aerospace, Hyderabad

We are proud to have supplied our high linear ramp rate Thermal Cyclic Test Chamber to BrahMos Aerospace, Hyderabad.

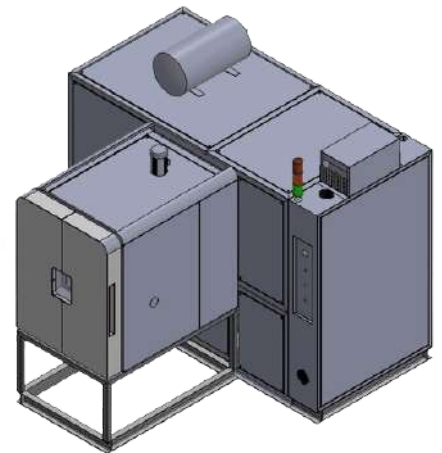
Key Specifications

Steel Mass Capacity: 50 kg

Heating Load: 1 kW

Ramp Rate: 10°C per minute

This advanced test chamber is tailored to meet the demanding thermal cycling requirements of aerospace applications.



Special Walk-In Climatic Test Chamber by MAK Control System

Designed for large specimens up to 1800 kg, this top-opening chamber offers rapid temperature change rates of 1°C/min with a range of -40°C to +180°C and 30% to 98% RH.

Featuring integrated pressure and temperature sensors, it ensures precise testing while maintaining a low noise level for a comfortable working environment.

Ideal for high-throughput testing, it enhances user efficiency and sample integrity.



Customized Walk-In Climatic Test Chamber for DRDL

- We supplied a Walk-In Climatic Test Chamber to DRDL for testing BrahMos missile steel structures.

- **Key Specifications**

Capacity: Handles 200 kg to 300 kg steel mass

Temperature Range: 90°C to -55°C within 45 seconds

Body Temperature Control: Ramp rate is achieved at the test specimen body temperature, not just air temperature

Sensor Integration: 32 sensors for precise temperature distribution and testing

- This chamber ensures accurate thermal testing for aerospace applications



Customized All-in-One Climatic Test Chamber for Isolator Testing

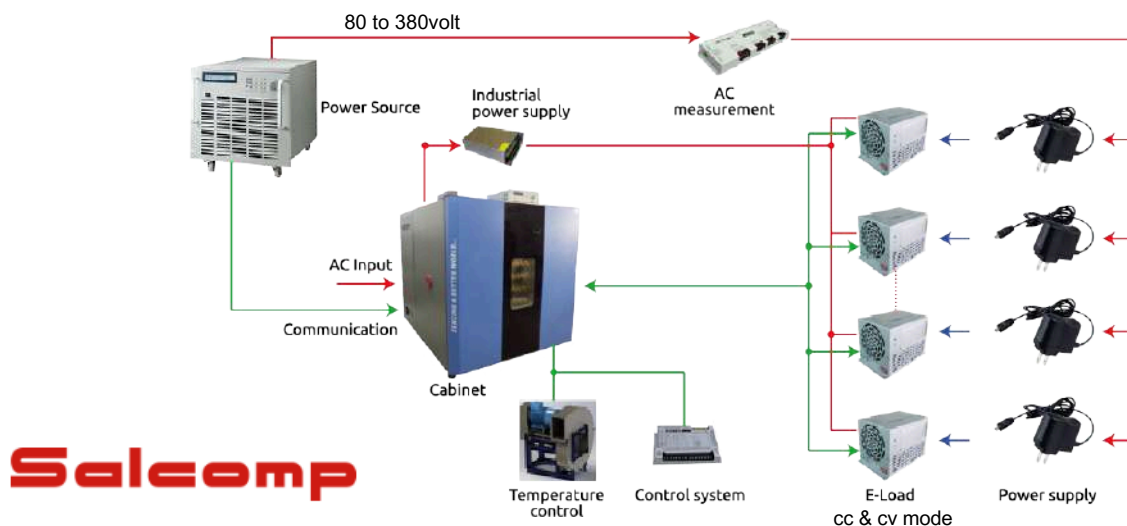
- We have developed a Custom Climatic Test Chamber specifically for performing multiple tests on transformer specimens using isolators. The chamber supports:

- - Solar Simulation
 - Rain Testing
 - Salt Spray Testing
 - Climatic Testing
 - High Voltage and Leakage Current Testing

- This versatile chamber ensures comprehensive evaluation of transformers under various environmental and electrical conditions.



Climatic Test Chamber, Designed specifically for aging cycle tests of Charger Units.



Salecomp

Revitalization of Asia's Largest Walk-In Altitude Test Chamber for Hindustan Aeronautics Limited

- We successfully revitalized Asia's largest walk-in altitude test chamber, originally supplied by an international company, extending its operational life beyond 10 years.
- After the compressor failure, **HAL** sought an economical and reliable solution. We conducted a thorough inspection and recommended upgrades that complied with government norms.
- The machine is now fully operational, and **HAL** is extremely satisfied with our refrigeration system upgrade, ensuring continued performance and reliability.



Before



After

- we recently completed a major modification for a thermal shock test chamber from another brand.



Before



After



TUV Test Lab in Pune Utilizing Our Advanced Test Chambers

We supplied a Altitude Test Chamber (capable of reaching 1 mbar) and a suite of other advanced testing chambers have completed a successful year at the MECF (MCCIA Electronic Cluster Foundation) and TUV Rheinland's state-of-the-art test facility in Pune. This facility, which supports high-profile customers including DRDO, ISRO, and leading automotive companies.



CEV in Chennai Utilizing Our Advanced Climatic Test Chamber

We recently supplied a Advanced Climatic Test Chamber the new cutting edge control system based on SESS Climatic S/W, Which makes it possible to manage and monitor the chamber from the on board panel and desktop. This line of chambers comes in both Model ETC- climatic (temperature and humidity) And Model – TC Thermostatic (temperature only)



Acoustic Test Chamber

An acoustic test chamber is a specialized environment designed to measure and analyze sound properties without interference from external noise. These chambers are used for various applications, including testing the acoustical performance of materials, measuring sound levels, and studying sound propagation characteristics.

Key Specifications:

- Soundproofing, Controlled Environment,
- Reverberation Time Measurement,
- Absorption and Transmission Loss ,
- Equipment, Applications

General Specifications:

- Size and Dimensions, Sound Isolation,
- Interior Treatment, Access and Safety
- Door and Window Design,
- Temperature and Humidity Control,
- Measurement Equipment



Tabletop Climatic Test Chamber

Compact Design:

- Tabletop design for space-saving convenience.
- Compact size with dimensions: 400 x 400 x 400 mm (inner size).

Advanced Technology:

- PLC (Programmable Logic Controller) based test chamber.
- Equipped with a 7-inch HMI (Human-Machine Interface).
- HMI allows storing test data for up to 100 days.
- Capability to create and save up to 100 test profiles.

Humidity Control:

- RH (Relative Humidity): 10% to 95%.
- Ensures precise humidity control for varied testing needs.



Wide Temperature Range:

- Temperature range: -40°C to 180°C.
- Allows testing across a broad range of temperature conditions.

SERVICE ACHIEVEMENTS & PROJECTS

DRDO

At CABS Bangalore (DRDO Sector), we successfully completed a PLC replacement for a HALT/HASS test chamber as part of a project carried out a few years ago.

HANIL AUTOMOTIVE

We successfully modified the PLC and control systems of an ACS Italy Climatic Test Chamber and a Salt Spray Test Chamber a few years ago.

3M

We successfully carried out PLC modifications and control system upgrades for a CTC make Climatic Test Chamber at 3M Company and Schneider Electric a few years ago.

TATA ADVANCED

We successfully upgraded the PLC and refrigeration systems of an imported ACS Italy Climatic Test Chamber at Tata Advanced Systems, Bangalore, a few years ago.

HAL

We successfully undertook the refrigeration system modification for one of the largest altitude test chambers. This critical project was nearly completed to the topmost level, earning appreciation from the HAL team for our excellent execution.

OTHERS

We have successfully completed numerous internal test chamber service projects over the years. The examples mentioned above are just a few highlights of our expertise and accomplishments.

We have successfully completed numerous internal test chamber service projects, including custom modifications for aerospace and automotive chambers and system upgrades for various industries, ensuring improved reliability and performance.

CONTROL SYSTEM



Make: Siemens **PLC:** S7-1200 **HMI:** 7-inch Weintek Optional
PC Software: SESS Climatic Test Viewer LabVIEW-based Software

Key Specifications

Internal Memory: 4 GB, **RAM Memory:** 1 GB, **Ethernet Ports:** 2 , RS 485 **Port:** 1

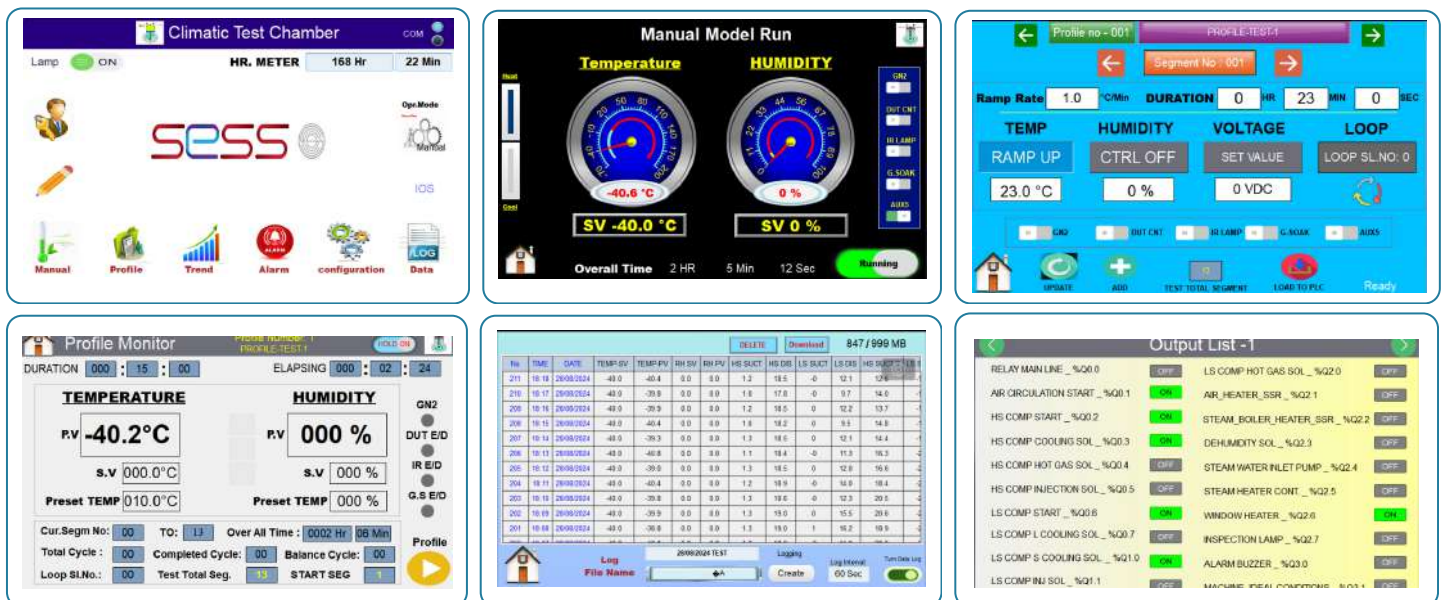
Data Logging: Up to 90 days, based on sampling duration (minimum 10 seconds)

Data Log Files: Downloadable via USB and PC, **USB Port:** 1

CMT Viewer: Accessible via PC and local network; mobile access through local network (customer-provided)

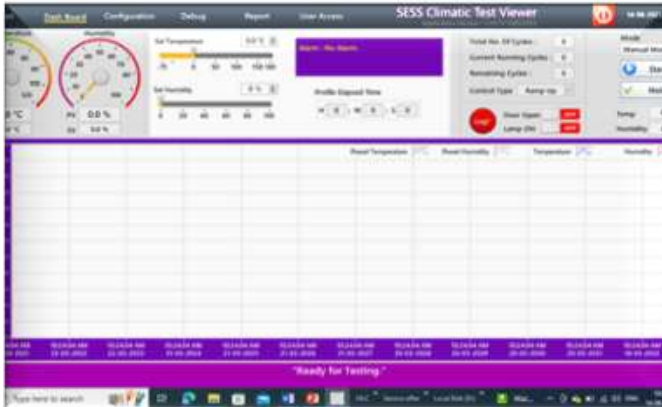
Profiles: Up to 100 profiles, each with up to maximum 100 segments per profile

HMI PAGE VIEW



PC SOFTWARE

START-UP



OPERATION IN MANUAL MODE



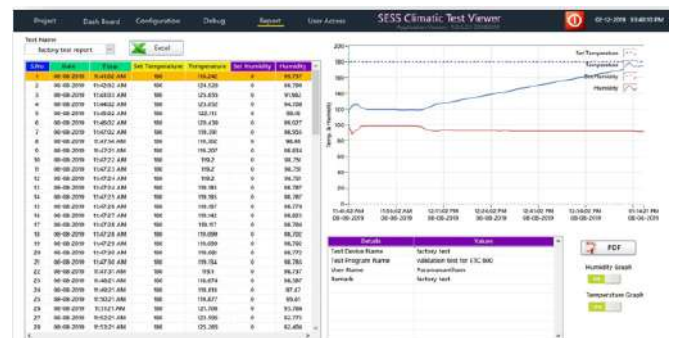
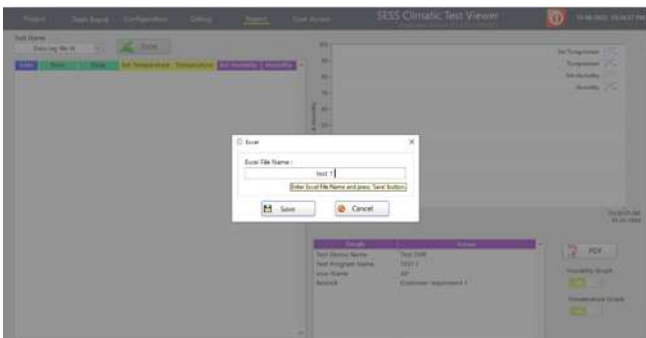
DI/DO SCREEN



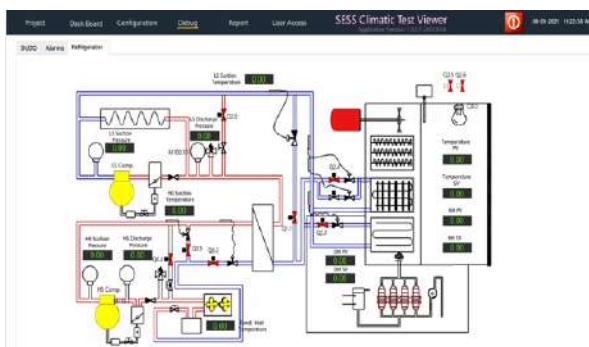
ALARM SCREEN



REPORT SCREEN



Refrigeration (or) Process and Instrumentation Diagram



HOW SESS HANDLES NEW PROJECTS

Initial Customer Inspection & Understanding

Objective: Understand the product details and customer application.

Activities:

- Site visit to inspect requirements.
- Gather application and test parameter details.



Customer SOR (Statement of Requirements)

Objective: Gather customer test parameters and specifications.

Activities:

- Receive SOR with test details.
- Clarify customer requirements for technical and commercial points.

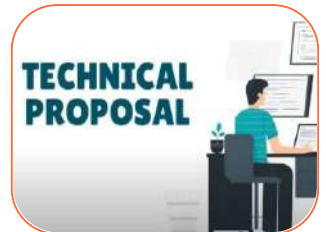


SESS Technical Offer & Commercial Proposal

Objective: Provide detailed technical and commercial offer.

Activities:

- Send technical offer with system details.
- Provide commercial pricing, lead times, and terms.

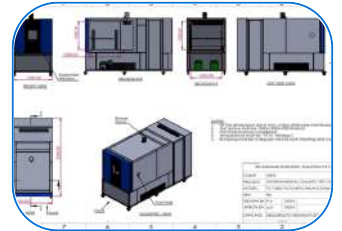


PO Received & GA (General Arrangement) Plan

Objective: Finalize the design after receiving PO.

Activities:

- Develop and send GA plan to customer for approval.
- Incorporate customer feedback and finalize GA.



Internal Process – Fabrication & QC

Objective: Begin chamber production and ensure quality control.

Activities:

- Sheet metal cutting (CNC).
- Fabrication of chamber structure.
- Powder coating and internal QC checks.



Pre-Assembly & Internal Testing

Objective: Assemble and test chamber components.

Activities:

- Pre-assemble mechanical, electrical, and refrigeration systems.
- Conduct internal tests and calibration for validation.



Pre-Delivery Inspection (PDI)

Objective: Allow customer to inspect the chamber before dispatch.

Activities:

- Schedule PDI at factory.
- Address any customer queries or concerns.
- Prepare final report post-PDI.



Dispatch & Site Preparation

Objective: Prepare for installation and dispatch the chamber.

Activities:

- Confirm site readiness and transport chamber.
- Ensure customer site meets installation requirements.



Installation & Final Testing

Objective: Install and validate the chamber at the customer site.

Activities:

- Send engineers for installation and testing.
- Perform final tests and ensure functionality.



Handover & Customer Sign-Off

Objective: Complete the project and hand over the chamber.

Activities:

- Provide user training and manuals.
- Final sign-off from customer after successful commissioning.



IN THE TUV TEST LAB, SOME OF OUR PRODUCTS WERE TESTED AS PART OF THE CE CERTIFICATION PROCESS.

Here are a few examples for your reference.



Contact Discharge



Air Discharge



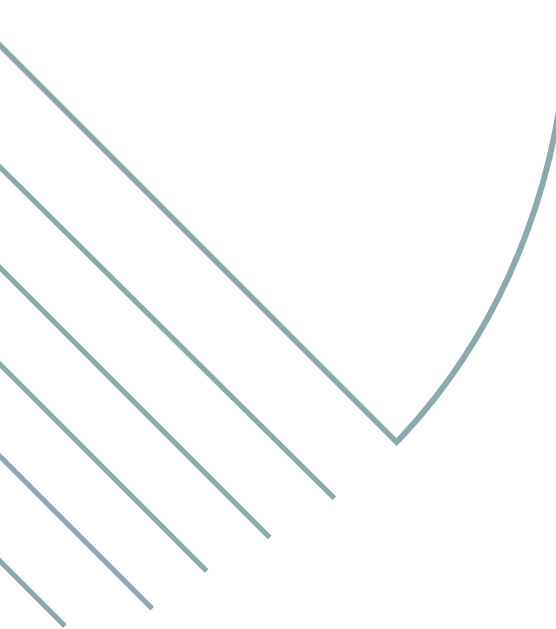
Conducted Emission Test Setup



Conducted RF Immunity Test Setup

CERTIFICATION





THANK YOU!

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